FREESE AND NICHOLS

Innovative approaches
Practical results
Outstanding service

WATER AND WASTEWATER RATE STUDY

Prepared for:

City of Lufkin

May 2018

Prepared by:

FREESE AND NICHOLS, INC.
4055 International Plaza, Suite 200
Fort Worth, Texas 76109
[redacted]

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FNI Project No.: LUF18860

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Appendix A: Historical and Projected Expenses for Water and Wastewater Fund

1.0 INTRODUCTION

The City of Lufkin provides retail water and wastewater services to customers within the City and outside the City limits (including the City of Burke). In addition, they also provide water supply to a number of wholesale customers. In November 2018, the City authorized Freese and Nichols, Inc. to perform a water and wastewater rate study. The purpose of the rate study is to develop a multi-year plan of rates that will provide sufficient revenue to implement needed capital improvements and allow the water and wastewater system to be self-supporting. FNI's methodology for performing this study is based on accepted industry standards and practices, specifically the American Water Works Association (AWWA) Manual 1 (M1) "Principles of Water Rates, Fees, and Charges", Seventh Edition.

The study included the following steps:

- Obtain data from City needed for the rate study.
- Review and analyze the data and develop projections of the operating expenses and revenues at current rates.
- Identify the revenue requirements for the water and wastewater system using the projected operating expenses and projected capital expenses in the next few years.
- Determine the impact of adjusting wholesale rates.
- Develop a rate model for the City's water and wastewater systems that projects the rates needed to provide sufficient annual revenue.
- Document the findings in a report.

This report describes the findings and recommendations of the water and wastewater rate study. The calculations are based on the City of Lufkin's Fiscal Year, which runs from October 1 through September 30.

2.0 WATER AND WASTEWATER SYSTEM

2.1 GENERAL DESCRIPTION

The City of Lufkin has a population of approximately 36,000 and owns and operates its own retail water and wastewater system. The City's water supply is from its own groundwater wells. The City owns and operates its own wastewater treatment plant. Lufkin has industrial customers that use significant amounts of water. In addition, Lufkin provides wholesale water supply to several cities and water supply corporations, as well as retail service to the City of Burke

The City's water distribution system consists of ## groundwater wells, ## elevated storage tanks (ESTs), # ground storage tanks (GSTs), and # pump stations. The City is divided into # pressure planes(?) with a total of ## miles of water lines ranging in diameter from 1-inch to 18-inches. Over the last several years, the City's average annual use has been approximately 7.4 MGD. This annual use is expected to increase with future population growth.

The City of Lufkin's wastewater service area covers approximately ## square miles. Within the service area, there are approximately ## miles of sewer lines that are owned and operated by the City. The City owns and operates a wastewater treatment plant (WWTP) located _____of the City. The wastewater collection system is primarily a gravity flow system that follows the major drainage basins of the service area. The wastewater lines range from # to ##-inches in diameter. There are currently ## lift stations in the wastewater collection system.

The City's water and wastewater customer base includes residential, commercial, and industrial users within the City limits, some retail customers outside the City, and several wholesale customers. In total, the City has approximately 15,600 metered connections. Most of the meters serve residential uses, and most are 5/8" or ¾" meters (93%), typical for residential users. Breakdowns of meter sizes and account types in the retail water system are provided in Table 2-1 and Table 2-2. This data is based on active meter data from 2017, and it is assumed that the current meter makeup is similar to 2018.

A review of FY14 through FY18 historical financial and billing data provided by the City showed the following general patterns:

- The combined water and wastewater revenue has averaged $16.9 million.
- Total annual water consumption by category generally follows the pattern of 43% by Residential (including Irrigation), 27% by Industrial, 24% by Commercial, and 6% by Wholesale.

Table 2-1: Meter Sizes in Water System

METER SIZE (inches)	% of Total Water Meters
5/8 or 3/4	93.3%
1	3.5%
2	2.5%
4	0.6%
6	0.1%

Table 2-2: Account Types in Water System

Location	Description	% of total Water Meters
Inside City	Residential	71.80%
	Commercial	11.02%
	Industrial	0.10%
	Irrigation	3.49%
Outside City	Residential	5.74%
	Commercial	0.22%
	Wholesale	0.03%
Burke	Residential	7.46%
	Commercial	0.13%

2.2 HISTORICAL AND PROJECTED WATER NEEDS AND WASTEWATER FLOWS

Table 2-3 shows historical data on population, number of connections, water production, metered water sales, and wastewater flows for Lufkin. The table shows nominal increases in population over the past five years. **Table 2-3** also shows that the percentage of nonrevenue water in recent years varies between 8 and 21 percent. Values of nonrevenue water above 12% are considered high for a retail water supplier. However, not all this nonrevenue water is unaccounted for. Some is used for authorized, beneficial purposes such as fire flow, line flushing, and irrigating city facilities such as parks. Currently the city estimates the amount used for these purposes. We recommend that the City meters as much of its unbilled authorized usage as feasible, particularly irrigation of city facilities. This will enable the City to better track how much water is actually being lost through leaks in their system. The City should continue to monitor nonrevenue water and perform detailed water audits in an effort to further reduce nonrevenue water. For this study, future projections of nonrevenue water will be 19 percent, based on

the average of the three highest years out of the last five years. Wastewater flow has consistently been 75% of the water production, and this is what has been used for projections of wastewater flow.

Table 2-3: Historical Population, Connections, Water Production, Water Sales, and Wastewater Flows

	Historical					*Budget*
	2013-14	2014-15	2015-16	2016-17	2017-18	*2018-19*
Population	35,923	35,988	35,933	35,855	36,106	*36,467*
Water Connections	15,486	15,503	15,570	15,547	15,624	*15,780*
Wastewater Connections	12,426	12,575	12,608	12,549	12,675	*12,802*
Persons per Water Connection	2.32	2.32	2.31	2.31	2.31	*2.31*
Potable Water Production (MG)						
Groundwater Production (MG)	2,684	2,697	2,725	2,599	2,840	*2,702*
Non-Revenue Water (%)	18.6%	8.1%	17.8%	18.6%	21.1%	*19.0%*
Metered Water (MG)	2,184	2,478	2,241	2,115	2,240	*2,189*
Wastewater Flow (MG)						
Total Wastewater Billed	1,576	1,669	1,589	1,655	1,667	*1,642*
Wastewater Flow at WWTP	1,937	2,190	2,165	1,951	1,888	*2,023*
Wastewater (% of Production)	72.2%	81.2%	79.4%	75.1%	66.5%	*74.9%*

Table 2-4 shows projected service population, water and sewer connections, water production, metered water sales, and wastewater flows for Lufkin. These projections are based on historical trends and input from City staff. These projected water production and use figures are for an average year. Water use will tend to be more in dry years and less in wet years. **Figure 2-1** shows the historical and projected number of water and wastewater connections. **Figure 2-2** shows the historical and projected water production and wastewater flows.

Key Assumptions for Projecting Water Production, Sales and Wastewater Flows:

- Growth in water meters: 1 percent per year based on historical trends
- Service area population: 1 percent per year increase based on historical trends
- Raw water production: 203 gallons per person per day based on the lowest three years out of the last five years. NOTE: this per capita calculation includes total water production for Lufkin and wholesale customers, but does not include the population of those wholesale customers. This creates an artificially high per capita. However, this per capita is suitable for the purposes of projecting water production.
- Metered water sales: 19 percent less than raw water production based on recent years

- Wastewater flow: 75 percent of raw water production based on historical data

Table 2-4: Projected Population, Connections, Water Production, Water Sales, and Wastewater Flows

	Budget	Projected									
Fiscal Year	***2018-19***	**2019-20**	**2020-21**	**2021-22**	**2022-23**	**2023-24**	**2024-25**	**2025-26**	**2026-27**	**2027-28**	**2028-29**
Population	*36,467*	36,832	37,200	37,572	37,948	38,327	38,710	39,097	39,488	39,883	40,282
Water Connections	*15,780*	15,938	16,097	16,258	16,421	16,585	16,750	16,918	17,087	17,258	17,431
Sewer Connections	*12,802*	12,930	13,059	13,190	13,322	13,455	13,590	13,726	13,863	14,002	14,142
Person per Connection	*2.31*	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31	2.31
Normal Year Potable Water Production (MG)	*2,702*	2,729	2,756	2,784	2,812	2,840	2,868	2,897	2,926	2,955	2,985
Non-Revenue Water (%)	*19%*	19%	19%	19%	19%	19%	19%	19%	19%	19%	19%
Metered Water Sales (MG)	*2,189*	2,210	2,232	2,255	2,278	2,300	2,323	2,347	2,370	2,394	2,418
Total Wastewater Billed	*1,642*	1,658	1,674	1,691	1,709	1,725	1,742	1,760	1,778	1,796	1,814
Wastewater Flows (MG)	*2,023*	2,043	2,063	2,084	2,105	2,126	2,147	2,169	2,191	2,212	2,235

MG = Million Gallons

4 way water for ~~AFP + Sterling~~ 20 MG/yr - Wholesale
Twin Disc. ~~Ind~~. Commercial
Power Plant 60 MG/yr RAW & Sewer
Huntington - 2 MG/~~yr~~ Wholesale
AFP + Sterling 2.6 MG/yr Industrial + Sewer

Figure 2-1: Historical and Projected Connections

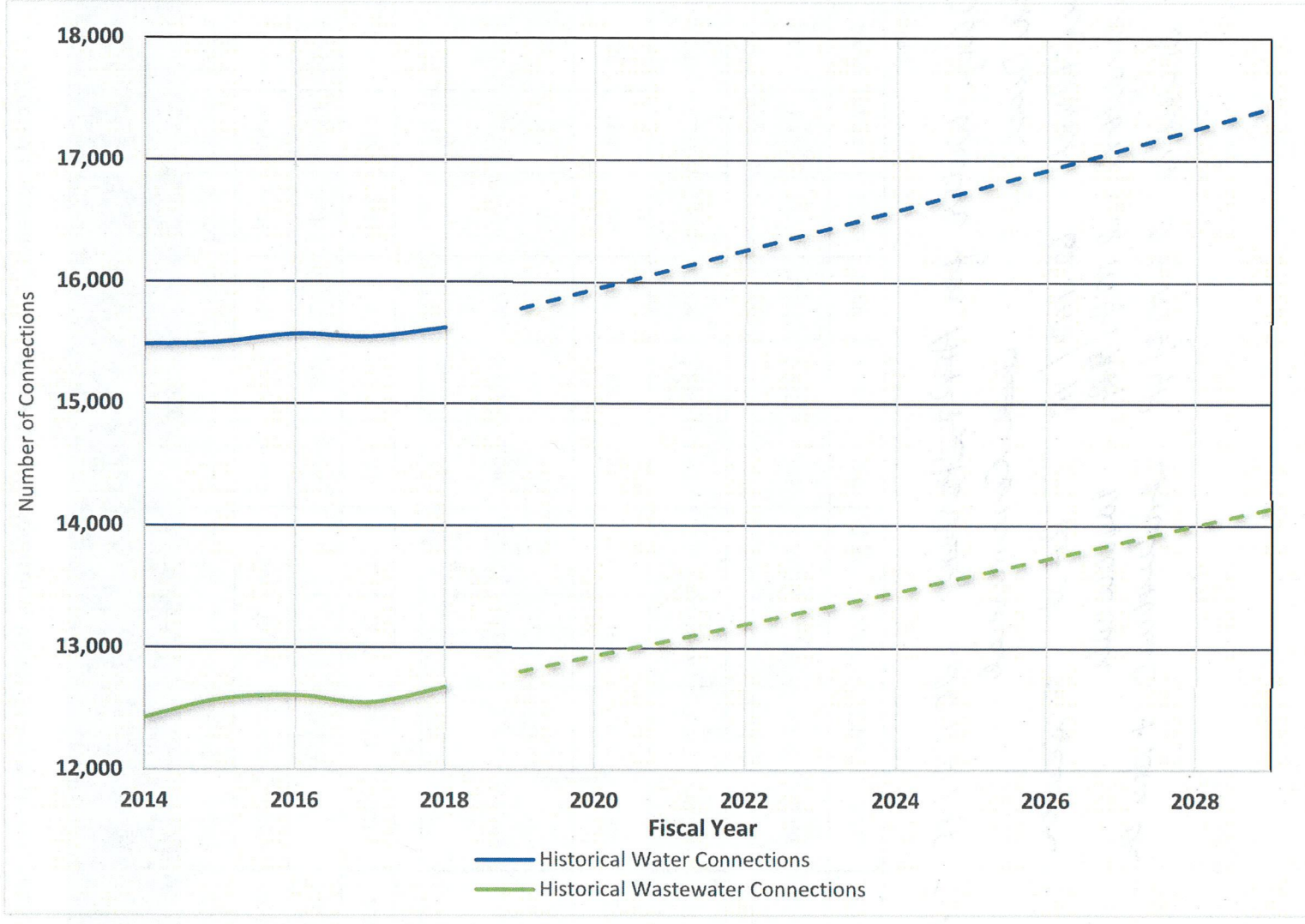
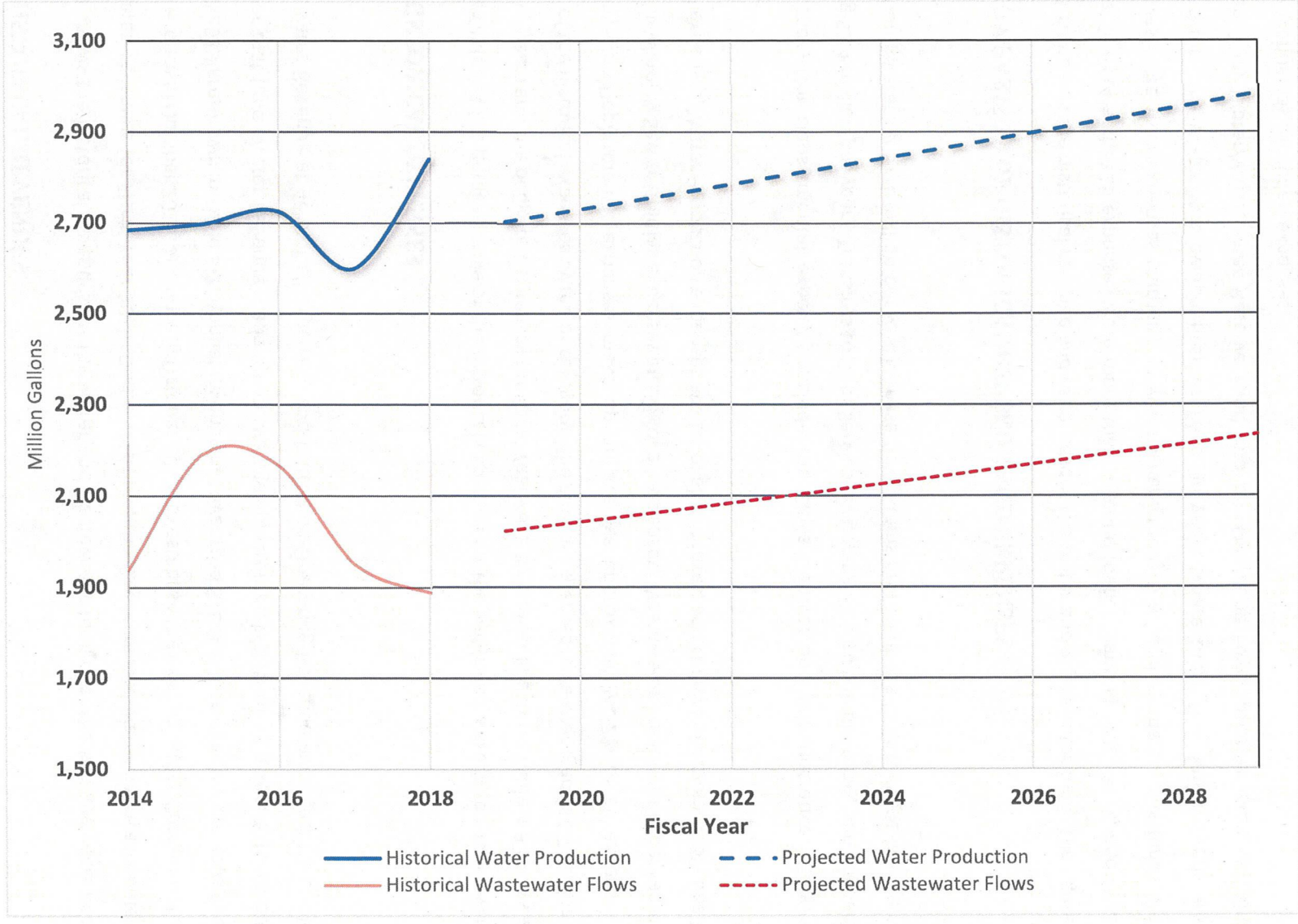


Figure 2-2: Historical and Projected Water Production and Wastewater Flow



3.0 HISTORICAL REVENUES AND EXPENSES

3.1 HISTORICAL REVENUES

Table 3-1 shows the historical revenues for the water and wastewater fund over the last five years and the budgeted revenues for Fiscal Year 2019. During this time, the City increased water and wastewater rates twice: once in October 2015, and once in October 2017 (to establish the existing rates). Current rates as of April 2019 are shown in **Table 3-2.** Annual revenue has averaged $16.9 million over the previous five years. Over that time, Water Revenue made up on average approximately 53% of all revenue, followed by Wastewater Revenue at 46% of all expenses, and lastly Non-Operating Revenue averaged 1% of all expenses.

3.2 HISTORICAL EXPENSES

Appendix Table A-1 is a detailed table of the historical expenses for the water and wastewater fund over the last five years and the budgeted expenses for Fiscal Year 2019 by budget line item. **Table 3-3** shows a summary of the historical expenses by major category. The annual expenses have averaged $16.7 million. Historically, Non-Departmental expenses make up on average approximately 52% of all expenses, followed by Water & Sewer Utilities expenses at 19% of all expenses, Wastewater Treatment expenses at 12% of all expenses, Water Production expenses at 11% of all expenses, and Utility Collections at 5% of all expenses.

For reference, Non-Departmental expenses include the City's General and Administrative Charges (averaging $2.4 million per year), Transfers to Debt Service (averaging $3.5 million per year), and Funded Depreciation (averaging $2.1 million per year). These three items represent the largest single line item expenses.

3.3 COMPARISON OF HISTORICAL REVENUE AND EXPENSES

Figure 3-1 compares the historical revenue and expenses for the water and wastewater fund. The figure shows that in FY14 and FY15 expenses were more than revenue (deficit), and in FY16, FY17, and FY18 revenue was more than expenses (surplus). In years where there is a surplus, the surplus funds are transferred into the Water/Wastewater Reserve Fund. In years where there is a deficit, funds are transferred from Water/Wastewater Reserve Fund into the Operating Fund to cover expenses. On average, revenue has exceeded expenses.

Table 3-1: Historical Revenue for the Water and Wastewater Fund

Category	Historical					*Budget*
	2013-14	2014-15	2015-16	2016-17	2017-18	*2018-19*
OPERATING REVENUE						
WATER REVENUE						
RESIDENTIAL						
Lufkin Res. Water Sales	$3,100,495	$3,318,695	$3,311,829	$3,194,842	$3,384,734	*$3,575,000*
Burke Area Water Sales	$508,651	$582,599	$566,391	$578,321	$604,643	*$592,000*
Lufkin Res. O/S C Limits	$361,222	$358,165	$384,511	$380,615	$405,440	*$399,000*
Irrigation Water Sales	$404,781	$441,488	$461,802	$402,271	$497,821	*$475,000*
Subtotal Residential	**$4,375,149**	**$4,700,946**	**$4,724,533**	**$4,556,048**	**$4,892,639**	***$5,041,000***
COMMERCIAL						
Lufkin Comm Water Sales	$2,186,935	$2,278,902	$2,428,839	$2,473,716	$2,525,360	*$2,545,000*
Burke Comm. Water Sales	$19,553	$20,966	$18,167	$15,666	$17,574	*$18,500*
Lufkin Comm. O/S C Limits	$34,670	$64,310	$52,401	$49,564	$51,291	*$50,000*
Subtotal Commercial	**$2,241,158**	**$2,364,179**	**$2,499,407**	**$2,538,946**	**$2,594,225**	***$2,613,500***
INDUSTRIAL SALES						
Lufkin Ind. Water Sales	$1,436,876	$1,311,721	$1,536,821	$1,322,471	$1,433,858	*$1,380,000*
Subtotal Industrial	**$1,436,876**	**$1,311,721**	**$1,536,821**	**$1,322,471**	**$1,433,858**	***$1,380,000***
WHOLESALE TREATED SALES						
Angelina FW Corp	$29,937	$24,772	$27,186	$27,134	$30,514	*$29,000*
City of Huntington	$120	$135	$135	$2,641	$135	*$135*
Redland Water Supply	$36,168	$37,907	$39,922	$44,992	$45,028	*$50,000*
Woodlawn WSC	$37,636	$68,024	$104,218	$97,730	$107,466	*$110,000*
City of Diboll	$173,777	$215,863	$186,918	$143,873	$211,322	*$175,000*
Subtotal Wholesale	**$277,639**	**$346,701**	**$358,417**	**$316,371**	**$394,465**	***$364,135***
SERVICE REVENUE						
Water Connections	$72,091	$66,539	$87,205	$87,497	$63,704	*$75,000*
Water Service Charge	$74,355	$89,808	$88,719	$81,921	$82,831	*$80,000*
Cash Over/Short	-$81	-$78	$3	-$73	-$192	*$100*
Subtotal Service Revenue	**$146,365**	**$156,269**	**$175,927**	**$169,345**	**$146,342**	***$155,100***
Kurth Water Sales	**$0**	**$0**	**$0**	**$0**	**$0**	***$0***
Subtotal Water Revenue	**$8,477,187**	**$8,879,817**	**$9,295,105**	**$8,903,181**	**$9,461,529**	***$9,553,735***
WASTEWATER REVENUE						
RESIDENTIAL	$2,767,412	$2,744,389	$2,824,574	$2,777,070	$2,800,205	*$2,880,000*
COMMERCIAL	$2,446,614	$2,556,181	$2,703,389	$2,793,645	$2,807,558	*$2,800,000*
INDUSTRIAL	$1,783,726	$1,840,748	$2,160,630	$1,977,756	$2,599,833	*$2,360,000*
SERVICE REVENUE						
Wastewater Connections	$27,406	$22,028	$19,638	$22,173	$41,444	*$22,000*
Pretreatment Testing	$56,153	$68,142	$56,460	$53,882	$63,559	*$60,000*
Septage Dumping	$53,020	$31,691	$39,857	$79,482	$159,948	*$90,000*
Cash Over/Short	$0	$0	$0	$0	$0	*$100*
Subtotal Service Revenue	**$136,578**	**$121,861**	**$115,955**	**$155,537**	**$264,951**	***$172,100***
Subtotal Wastewater Revenue	**$7,134,330**	**$7,263,179**	**$7,804,548**	**$7,704,008**	**$8,472,547**	***$8,212,100***
Total Operating Revenue	**$15,611,518**	**$16,142,996**	**$17,099,653**	**$16,607,188**	**$17,934,077**	**$17,765,835**
NON-OPERATING REVENUE						
Charges for Services	$9,811	$9,132	$7,742	$13,885	$12,007	*$9,000*
Interest Revenue	$16,493	$11,965	$30,845	$29,957	$37,444	*$23,500*

Category	Historical					Budget
	2013-14	2014-15	2015-16	2016-17	2017-18	*2018-19*
Inter/Intra-Gov. Revenue	$119,760	$119,760	$119,803	$123,572	$129,186	*$131,408*
Other Revenue	$78,288	$49,819	$42,481	$83,887	$2,893	*$16,700*
Solid Waste & Recycling	$1	$0	$500	$100	$1,900	*$0*
Total Non-Operating Rev.	**$224,353**	**$190,676**	**$201,371**	**$251,402**	**$183,430**	***$180,608***
TOTAL REVENUE	**$15,835,871**	**$16,333,672**	**$17,301,024**	**$16,858,590**	**$18,117,507**	***$17,946,443***

Table 3-2: City of Lufkin Monthly Water & Wastewater Rates (Effective October 2017)

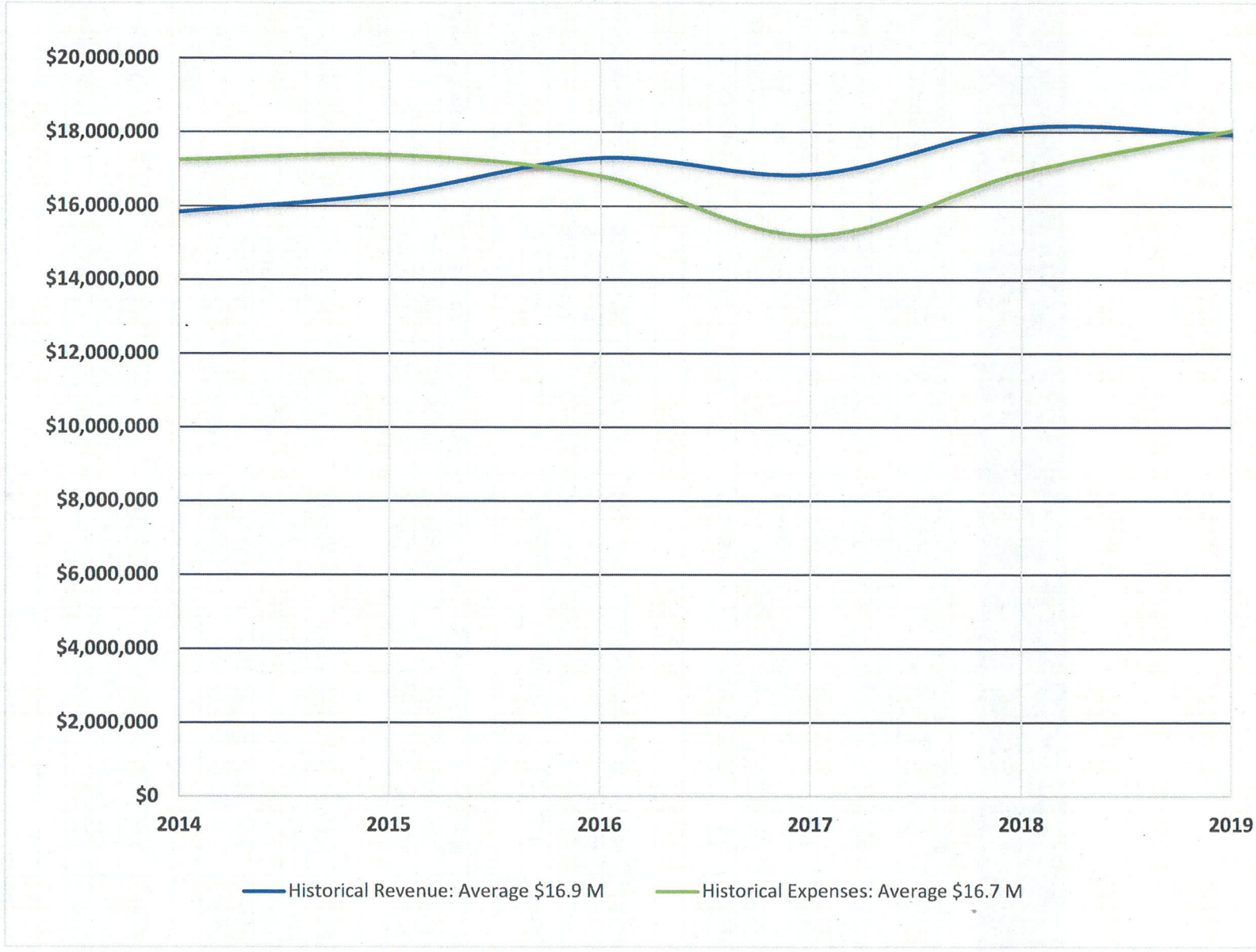
Current Water Rates		
Residential/Irrigation		
Service Charge (includes 2,000 gallons)	$13.26	
Volume Charge, Tier 1 (2,001-10,000 gallons)	$3.40	per 1,000 gal
Volume Charge, Tier 2 (over 10,000 gallons)	$3.93	per 1,000 gal
Commercial		
Service Charge (includes 2,000 gallons)	$13.26	
Volume Charge, Tier 1 (2,001-50,000 gallons)	$3.40	per 1,000 gal
Volume Charge, Tier 2 (50,001 gallons and above)	$3.93	per 1,000 gal
Industrial Inside City Limits		
Service Charge (includes 2,000 gallons)	$313.76	
Volume Charge, Tier 1 (over 2,000 gallons)	$2.22	per 1,000 gal
Current Wastewater Rates		
Residential		
Service Charge (includes 2,000 gallons)	$11.99	
Volume Charge (2,001-20,000 gallons, winter average)	$3.88	per 1,000 gal
Restaurant		
Service Charge (includes 2,000 gallons)	$11.99	
Volume Charge (over 2,000 gallons, winter average)	$8.46	per 1,000 gal
Commercial		
Service Charge (includes 2,000 gallons)	$11.99	
Volume Charge (over 2,000 gallons, winter average)	$4.29	per 1,000 gal
Industrial		
Service Charge (includes no volume)	$11.99	
Volume Charge	$2.36	per 1,000 gal
	$0.0040469	mg/l per 1,000 gallons
	$0.0022260	mg/l per 1,000 gallons

Note: Water rates for customers Outside City Limits are 1.5 times the Inside City Limits rates.
Wastewater rates for customers Outside City Limits are 2 times the Inside City Limits rates.

Table 3-3: Historical Expenses for the Water and Wastewater Fund

Expense Category	Historical					*Budget*
	2013-14	2014-15	2015-16	2016-17	2017-18	*2018-19*
Utility Collections						
Billing & Collections	$506,275	$475,069	$569,919	$585,815	$573,300	*$578,899*
Meter Reading	$269,932	$226,923	$287,316	$284,904	$271,769	*$297,155*
Utility Collections Subtotal	**$776,207**	**$701,992**	**$857,235**	**$870,719**	**$845,069**	***$876,054***
Wastewater Treatment						
Administration	$833,572	$755,587	$835,457	$822,332	$719,413	*$849,123*
Operations	$402,841	$388,849	$460,526	$430,399	$485,906	*$466,431*
Maintenance	$549,407	$537,068	$661,128	$677,934	$604,308	*$620,628*
Laboratory	$200,334	$197,702	$268,017	$261,968	$219,699	*$240,728*
Wastewater Treatment Subtotal	**$1,986,154**	**$1,879,206**	**$2,225,128**	**$2,192,633**	**$2,029,326**	***$2,176,910***
Water Production						
Operations	$1,696,011	$1,730,395	$1,705,829	$1,780,600	$1,763,499	*$1,946,380*
Maintenance	$142,096	$150,399	$136,120	$158,294	$150,991	*$160,441*
Water Production Subtotal	**$1,838,107**	**$1,880,794**	**$1,841,949**	**$1,938,894**	**$1,914,490**	***$2,106,821***
Water & Sewer Utilities						
Administration	$467,540	$398,389	$448,616	$483,579	$517,376	*$554,709*
Water Utilities	$1,146,231	$1,093,377	$1,270,007	$1,239,768	$1,266,025	*$1,406,385*
Sewer Utilities	$1,618,328	$1,382,012	$1,595,119	$1,652,867	$1,556,882	*$1,653,053*
Water & Sewer Utilities Subtotal	**$3,232,099**	**$2,873,778**	**$3,313,742**	**$3,376,214**	**$3,340,283**	***$3,614,147***
Non-Departmental						
Administration						
Gen. & Admin. Charges	$2,224,072	$2,310,725	$2,385,669	$2,440,991	$2,641,151	*$2,788,422*
Funded Depreciation	$2,007,130	$2,056,085	$2,110,278	$2,110,278	$2,107,905	*$2,079,977*
Depreciation Expense	$2,359,022	$2,356,001	$2,355,686	$2,387,519	$0	*$0*
Transfers-General Fund	$409,000	$100,000	$368,928	$565,518	$100,000	*$100,000*
Transfers-Gr Health Ins	$500,000	$0	$0	$0	$0	*$0*
Transfers-GO Debt Service	$3,431,981	$3,771,795	$3,129,448	$3,778,005	$3,546,645	*$3,759,552*
Fixed Asset	-$1,898,258	-$841,338	-$2,430,643	-$4,860,833	$0	*$0*
All Other Administration	$387,764	$293,624	$657,236	$404,871	$374,807	*$541,505*
Non-Departmental Subtotal	**$9,420,711**	**$10,046,892**	**$8,576,602**	**$6,826,349**	**$8,770,508**	***$9,269,456***
Total Operating Expenses	**$17,253,278**	**$17,382,662**	**$16,814,656**	**$15,204,809**	**$16,899,676**	***$18,043,388***
Total Revenue from Table 3-1	**$15,835,871**	**$16,333,672**	**$17,301,024**	**$16,858,590**	**$18,117,507**	***$17,946,443***
Deficit (-) or Surplus (+)	**-$1,417,407**	**-$1,048,990**	**$486,368**	**$1,653,781**	**$1,217,831**	***-$96,945***

Figure 3-1: Historical Water and Wastewater Revenue and Expenses



4.0 WATER AND WASTEWATER FUND RATE ANALYSIS

4.1 PROJECTED REVENUES AT CURRENT RATES

Table 4-1 shows projected revenues at current rates for the water and wastewater fund. Projected Revenue for FY20 is approximately $18.1 million. If held to current rates, revenue is anticipated to increase by approximately 1 percent per year. This 1% increase generally follows the trend in increased population, connections, and water/wastewater use. **Figure 4-1** shows the historical water and wastewater revenues and projected revenues at current rates. The key assumptions in developing the projected revenues are provided in **Table 4-2.**

4.2 PROJECTED EXPENSES

Appendix Table A-2 shows projected expenses by detailed category for the water and wastewater fund. **Appendix Table A-3** outlines the basis for developing the projections for each detailed category. **Table 4-3** is a summary by major category of the projected expenses for the water and wastewater fund. Projected expenses for FY20 are approximately $18 million. In future years, expenses are projected to increase at a modest rate, primarily due to inflation and other escalators except for years when existing debt drops off. Debt service currently makes up approximately 20% of the overall expenses ($3.8 million out of the $18 million budgeted in FY19) and is forecasted to average around that same percentage between FY19 and FY29.

Over the next 10 years, the City is anticipated to undertake approximately $15 million in improvements to its water and wastewater system. Half will be needed in FY20 and the other half in FY22. About three-fourths of the funds will be for water projects and the remaining one-fourth for wastewater. These improvements are necessary to provide for future growth, replace aging infrastructure, maintain reliable service, keep facilities in compliance with new regulations, and improve efficiency. Debt service for each year's bond issue was calculated using an interest rate of 3.0 percent, 20-year repayment terms, and a yearly inflation rate of 3 percent applied to the present-day capital cost of CIP projects that will be constructed in future years. No coverage requirement was assumed since the City plans to use General Obligation Bonds which don't require coverage.

Figure 4-2 shows the historical and projected expenses for the water and wastewater fund, and **Figure 4-3** compares the total revenue and expenses, both historical and projected values, and shows that revenues will fall short of expenses after FY21 unless the City increases its water and wastewater rates.

Table 4-1: Projected Revenue at Current Rates for the Water and Wastewater Fund

	Budget	Projected									
	[redacted]	[redacted]20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29
Projected Water Connections	*15,780*	15,938	16,097	16,258	16,421	16,585	16,750	16,918	17,087	17,258	17,431
Projected Metered Water (MG) (billed & unbilled)	*2,189*	2,210	2,232	2,255	2,278	2,300	2,323	2,347	2,370	2,394	2,418
Metered Water Sales (MG) (billed)	*2,189*	*2,210*	*2,232*	*2,255*	*2,278*	*2,300*	*2,323*	*2,347*	*2,370*	*2,394*	*2,418*
Projected Sewer Connections	*12,802*	12,930	13,059	13,190	13,322	13,455	13,590	13,726	13,863	14,002	14,142
Projected Wastewater billed (MG)	*1,642*	1,658	1,674	1,691	1,709	1,725	1,742	1,760	1,778	1,796	1,814
OPERATING REVENUE											
WATER REVENUE											
Service Demand Charge		$2,536,009	$2,561,347	$2,586,961	$2,612,850	$2,638,945	$2,665,316	$2,691,962	$2,718,884	$2,746,081	$2,773,553
Volume Charge		$6,784,700	$6,852,240	$6,922,850	$6,993,460	$7,061,000	$7,131,610	$7,205,290	$7,275,900	$7,349,580	$7,423,260
Subtotal	***$9,034,500***	**$9,320,709**	**$9,413,587**	**$9,509,811**	**$9,606,310**	**$9,699,945**	**$9,796,926**	**$9,897,252**	**$9,994,784**	**$10,095,661**	**$10,196,813**
WASTEWATER REVENUE											
Service Demand Charge		$1,860,368	$1,878,929	$1,897,777	$1,916,769	$1,935,905	$1,955,329	$1,974,897	$1,994,608	$2,014,608	$2,034,751
Volume Charge		$5,935,640	$5,992,920	$6,053,780	$6,118,220	$6,175,500	$6,236,360	$6,300,800	$6,365,240	$6,429,680	$6,494,120
Subtotal	***$8,040,000***	**$7,796,008**	**$7,871,849**	**$7,951,557**	**$8,034,989**	**$8,111,405**	**$8,191,689**	**$8,275,697**	**$8,359,848**	**$8,444,288**	**$8,528,871**
WHOLESALE TREATED SALES											
Angelina FW Corp	*$29,000*	$29,000	$29,000	$29,000	$29,000	$29,000	$29,000	$29,000	$29,000	$29,000	$29,000
4-Way SUD	*$115,500*	$115,800	$116,100	$116,400	$116,700	$117,100	$117,500	$117,900	$118,300	$118,700	$119,100
Redland Water Supply	*$50,000*	$50,000	$50,000	$50,000	$50,000	$50,000	$50,000	$50,000	$50,000	$50,000	$50,000
Woodlawn WSC	*$110,000*	$110,110	$110,220	$110,330	$110,440	$110,550	$110,661	$110,772	$110,883	$110,994	$111,105
City of Diboll	*$175,000*	$175,525	$176,052	$176,580	$177,110	$177,641	$178,174	$178,709	$179,245	$179,783	$180,322
SUBTOTAL WHOLESALE	*$0*	*$0*	*$0*	*$0*	*$0*	*$0*	*$0*	*$0*	*$0*	*$0*	*$0*

SERVICE REVENUE											
Water Connections	*$75,000*	$75,000	$75,000	$75,000	$75,000	$75,000	$75,000	$75,000	$75,000	$75,000	$75,000
Water Service Charge	*$80,000*	$80,000	$80,000	$80,000	$80,000	$80,000	$80,000	$80,000	$80,000	$80,000	$80,000
Wastewater Connections	*$22,000*	$22,000	$22,000	$22,000	$22,000	$22,000	$22,000	$22,000	$22,000	$22,000	$22,000
Pretreatment Testing	*$60,000*	$60,000	$60,000	$60,000	$60,000	$60,000	$60,000	$60,000	$60,000	$60,000	$60,000
Septage Dumping	*$90,000*	$90,000	$90,000	$90,000	$90,000	$90,000	$90,000	$90,000	$90,000	$90,000	$90,000
Cash Over/Short	*$200*										
SUBTOTAL SERVICE REVENUE	***$327,200***	***$327,000***	***$327,000***	***$327,000***	***$327,000***	***$327,000***	***$327,000***	***$327,000***	***$327,000***	***$327,000***	***$327,000***
Subtotal	***$806,700***	***$807,435***	***$808,372***	***$809,310***	***$810,250***	***$811,291***	***$812,335***	***$813,381***	***$814,428***	***$815,477***	***$816,527***
TOTAL OPERATING REVENUE	***$17,881,200***	**$17,924,152**	**$18,093,808**	**$18,270,678**	**$18,451,549**	**$18,622,641**	**$18,800,950**	**$18,986,330**	**$19,169,060**	**$19,355,426**	**$19,542,211**
NON OPERATING REVENUE											
Charges for Services	*$9,000*	$9,000	$9,000	$9,000	$9,000	$9,000	$9,000	$9,000	$9,000	$9,000	$9,000
Interest Revenue	*$23,500*	$23,500	$23,500	$23,500	$23,500	$23,500	$23,500	$23,500	$23,500	$23,500	$23,500
Inter/Intra-Gov. Revenue	*$131,408*	$132,722	$134,050	$135,391	$136,745	$138,113	$139,494	$140,889	$142,298	$143,721	$145,159
Other Revenue	*$16,700*	$16,700	$16,700	$16,700	$16,700	$16,700	$16,700	$16,700	$16,700	$16,700	$16,700
Solid Waste & Recycling	*$0*	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
TOTAL NON OPERATING REVENUE	***$180,608***	**$181,922**	**$183,250**	**$184,591**	**$185,945**	**$187,313**	**$188,694**	**$190,089**	**$191,498**	**$192,921**	**$194,359**
TOTAL REVENUE	***$18,061,808***	**$18,106,074**	**$18,277,058**	**$18,455,269**	**$18,637,494**	**$18,809,954**	**$18,989,644**	**$19,176,419**	**$19,360,558**	**$19,548,347**	**$19,736,570**

MG = Million Gallons

Table 4-2: Assumptions in Developing Projected Revenues

OPERATING REVENUE Category	Basis for projections
WATER REVENUE	
Service Demand Charge	
Volume Charge	
WASTEWATER REVENUE	
Service Demand Charge	
Volume Charge	
WHOLESALE TREATED SALES	
Angelina FW Corp	Increase at same rate as State Water Plan demand projections, 0.0% per year
4-Way SUD	Increase at same rate as State Water Plan demand projections, 0.3% per year
Redland Water Supply	Increase at same rate as State Water Plan demand projections, 0.0% per year
Woodlawn WSC	Increase at same rate as State Water Plan demand projections, 0.1% per year
City of Diboll	Increase at same rate as State Water Plan demand projections, 0.3% per year
SERVICE REVENUE	
Water Connections	Hold constant at 2019 budget. Based on current rates/fees so will not go up until rates/fees increase.
Water Service Charge	Hold constant at 2019 budget. Based on current rates/fees so will not go up until rates/fees increase.
Wastewater Connections	Hold constant at 2019 budget. Based on current rates/fees so will not go up until rates/fees increase.
Pretreatment Testing	Hold constant at 2019 budget. Based on current rates/fees so will not go up until rates/fees increase.
Septage Dumping	Hold constant at 2019 budget. Based on current rates/fees so will not go up until rates/fees increase.
Cash Over/Short	
NON OPERATING REVENUE	
Charges for Services	Hold constant at 2019 budget. Based on current rates/fees so will not go up until rates/fees increase.
Interest Revenue	Hold constant at 2019 budget. Based on current rates/fees so will not go up until rates/fees increase.
Inter/Intra-Gov. Revenue	Begin with 2019 budget amount and increase by 1.0% /yr
Other Revenue	Hold constant at 2019 budget. Based on current rates/fees so will not go up until rates/fees increase.
Solid Waste & Recycling	Hold constant at 2019 budget. Based on current rates/fees so will not go up until rates/fees increase.

Figure 4-1: Historical and Projected Water and Wastewater Revenue

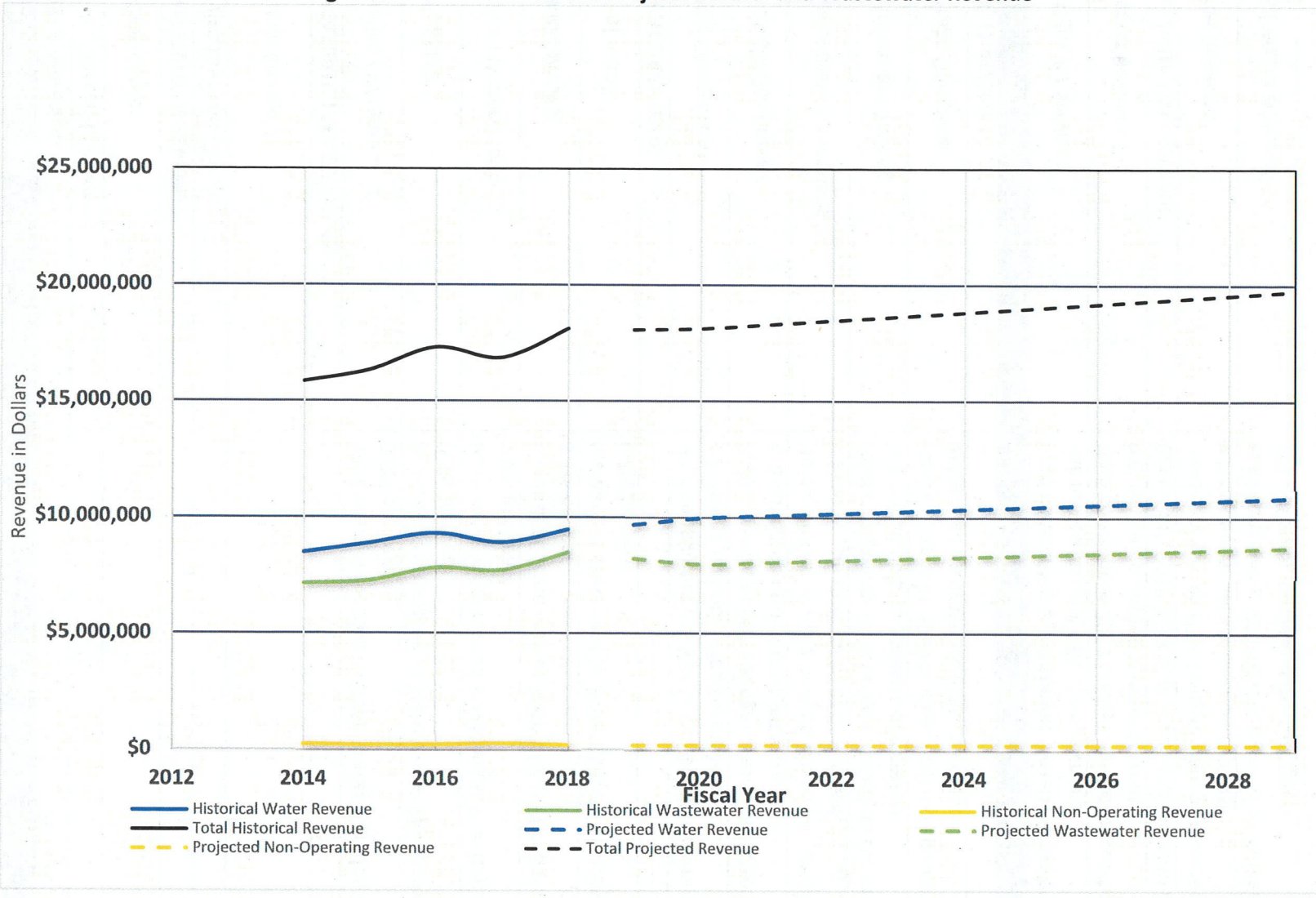


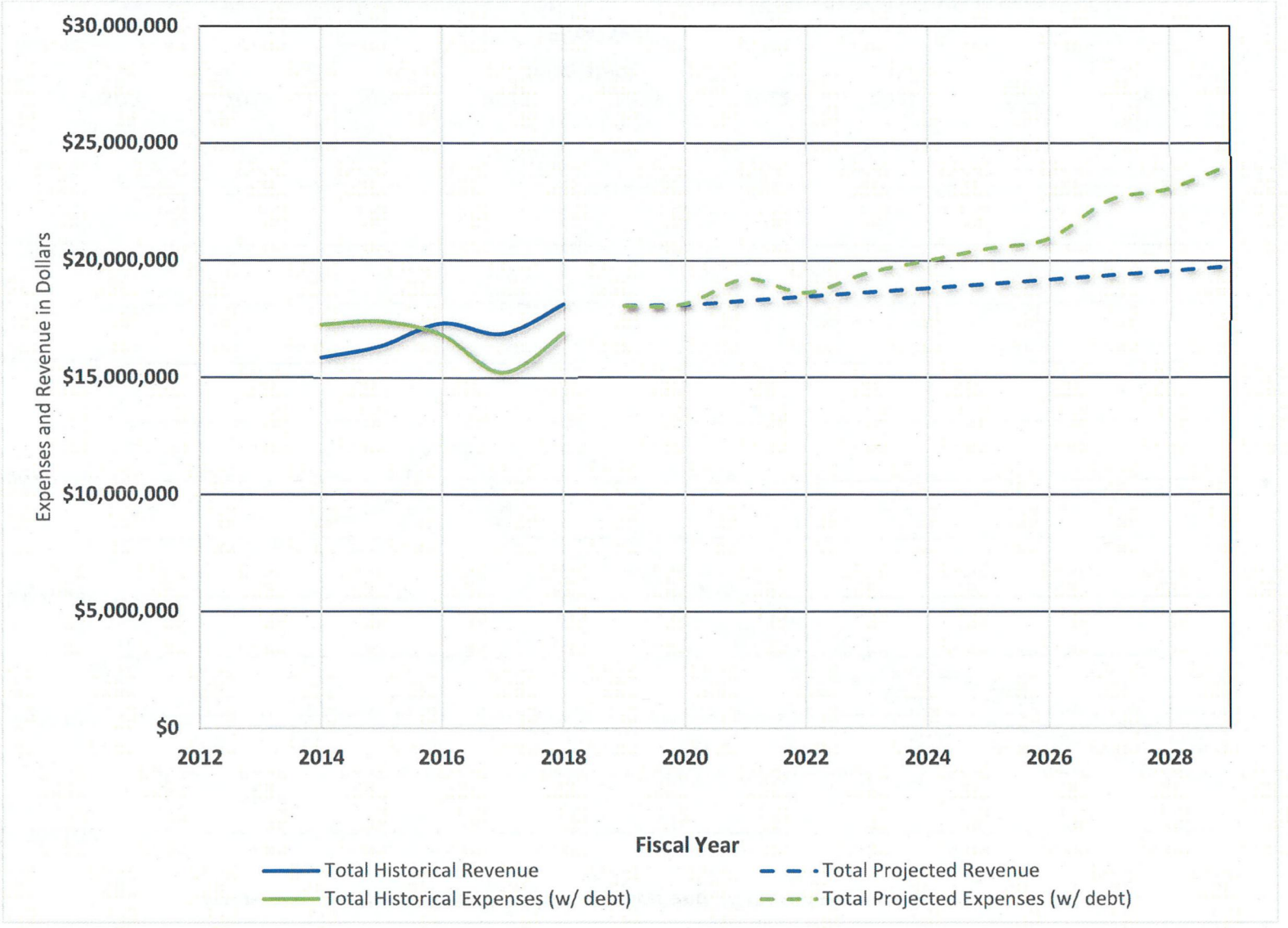
Table 4-3: Projected Expenses for the Water and Wastewater Fund, by Major Category

	Budget	Projected Expenses by Fiscal Year									
	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29
WATER SYSTEM											
Salaries & Benefits	$1,384,962	$1,433,952	$1,484,141	$1,536,086	$1,589,849	$1,645,493	$1,703,085	$1,762,693	$1,824,387	$1,888,241	$1,954,328
Supplies	$453,223	$467,324	$486,237	$506,016	$526,697	$548,325	$570,943	$594,607	$619,360	$645,256	$672,356
Repairs	$711,156	$685,072	$694,157	$703,708	$713,509	$723,584	$733,959	$744,896	$756,189	$767,873	$795,531
Misc. Service	$574,700	$555,427	$599,030	$646,658	$698,707	$755,614	$817,860	$885,981	$960,562	$1,042,254	$1,131,777
Utilities	$1,300,300	$1,233,508	$1,266,813	$1,301,361	$1,336,715	$1,372,893	$1,409,914	$1,448,295	$1,487,571	$1,527,763	$1,569,415
Chemicals	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Bad Debt	$75,000	$76,500	$78,030	$79,591	$81,183	$82,806	$84,462	$86,151	$87,874	$89,632	$91,424
G&A Charges	$1,444,211	$1,507,451	$1,573,519	$1,642,542	$1,714,653	$1,789,991	$1,868,702	$1,950,936	$2,036,852	$2,126,615	$2,220,399
Funded Depreciation	$1,039,989	$1,050,389	$1,060,893	$1,071,502	$1,082,217	$1,093,039	$1,103,969	$1,115,009	$1,126,159	$1,137,420	$1,148,794
Depreciation	$0	$1,203,600	$1,227,672	$1,252,226	$1,277,270	$1,302,816	$1,328,872	$1,355,450	$1,382,559	$1,410,210	$1,438,414
Fixed Assets	$0	-$1,280,100	-$1,305,702	-$1,331,816	-$1,358,453	-$1,385,622	-$1,413,334	-$1,441,601	-$1,470,433	-$1,499,841	-$1,529,838
Existing Debt Service	$1,879,776	$1,789,376	$1,823,664	$1,283,839	$1,175,701	$1,170,758	$1,147,279	$1,062,868	$1,571,397	$1,496,778	$1,649,680
Planned Debt Service	$0	$0	$385,650	$385,650	$786,881	$786,881	$786,881	$786,881	$786,881	$786,881	$786,881
TOTAL WATER SYSTEM	**$8,863,316**	**$8,722,498**	**$9,374,102**	**$9,077,360**	**$9,624,927**	**$9,886,575**	**$10,142,590**	**$10,352,162**	**$11,169,356**	**$11,419,079**	**$11,929,159**
SEWER SYSTEM											
Salaries & Benefits	$2,434,662	$2,520,393	$2,608,607	$2,699,908	$2,794,404	$2,892,209	$2,993,435	$3,098,205	$3,206,643	$3,318,875	$3,435,036
Supplies	$523,243	$553,281	$575,445	$598,830	$623,512	$649,578	$677,115	$706,222	$736,999	$769,558	$804,015
Repairs	$595,231	$625,459	$636,115	$647,233	$658,696	$670,531	$682,767	$695,573	$708,843	$722,615	$741,111
Misc. Service	$1,187,962	$1,226,029	$1,266,196	$1,308,638	$1,353,548	$1,401,138	$1,451,644	$1,505,331	$1,562,485	$1,623,426	$1,688,511
Utilities	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Chemicals	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Bed Debt	$75,000	$76,500	$78,030	$79,591	$81,183	$82,806	$84,462	$86,151	$87,874	$89,632	$91,424
G&A Charges	$1,444,211	$1,507,451	$1,573,519	$1,642,542	$1,714,653	$1,789,991	$1,868,702	$1,950,936	$2,036,852	$2,126,615	$2,220,399

	Budget	Projected Expenses by Fiscal Year									
	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29
Funded Depreciation	*$1,039,989*	$1,050,389	$1,060,893	$1,071,502	$1,082,217	$1,093,039	$1,103,969	$1,115,009	$1,126,159	$1,137,420	$1,148,794
Depreciation	*$0*	$1,203,600	$1,227,672	$1,252,226	$1,277,270	$1,302,816	$1,328,872	$1,355,450	$1,382,559	$1,410,210	$1,438,414
Fixed Assets	*$0*	-$1,280,100	-$1,305,702	-$1,331,816	-$1,358,453	-$1,385,622	-$1,413,334	-$1,441,601	-$1,470,433	-$1,499,841	-$1,529,838
Existing Debt Service	*$1,879,776*	$1,789,376	$1,823,664	$1,283,839	$1,175,701	$1,170,758	$1,147,279	$1,062,868	$1,571,397	$1,496,778	$1,649,680
Planned Debt Service	*$0*	$0	$128,550	$128,550	$262,294	$262,294	$262,294	$262,294	$262,294	$262,294	$262,294
TOTAL SEWER SYSTEM	***$9,180,073***	***$9,272,377***	***$9,672,987***	***$9,381,040***	***$9,665,023***	***$9,929,535***	***$10,187,203***	***$10,396,434***	***$11,211,670***	***$11,457,579***	***$11,949,838***
TOTAL WATER AND SEWER	***$18,043,388***	***$17,994,875***	***$19,047,089***	***$18,458,399***	***$19,289,950***	***$19,816,110***	***$20,329,793***	***$20,748,596***	***$22,381,025***	***$22,876,657***	***$23,878,996***

Figure 4-2: Historical and Projected Water and Wastewater Expenses

Figure 4-3: Historical and Projected Water and Wastewater Expenses and Revenue (at current rates)



APPENDIX A

HISTORICAL AND PROJECTED EXPENSES FOR WATER AND WASTEWATER FUND

Table A-1: Historical Expenses for the Water and Wastewater Fund

	Historical Expenses by Fiscal Year					*Budgeted*
	2013-14	**2014-15**	**2015-16**	**2016-17**	**2017-18**	***2018-19***
Utility Collections						
Billing & Collections						
Salaries & Wages	$245,958	$254,711	$261,490	$261,575	$262,814	*$261,419*
Benefits	$108,682	$66,607	$127,999	$139,055	$110,075	*$117,765*
Supplies	$75,995	$73,757	$83,680	$76,315	$73,048	*$79,760*
Repairs	$2,687	$2,586	$4,599	$6,355	$5,268	*$9,330*
Misc. Service	$72,953	$77,408	$92,151	$102,515	$109,195	*$110,625*
Machinery & Equipment	$0	$0	$0	$0	$12,900	*$0*
SUBTOTAL Billing & Collections	**$506,275**	**$475,069**	**$569,919**	**$585,815**	**$573,300**	***$578,899***
Meter Reading						
Salaries & Wages	$152,794	$150,099	$166,483	$159,447	$153,439	*$155,384*
Benefits	$74,049	$39,747	$89,396	$89,474	$73,381	*$73,412*
Supplies	$21,190	$18,655	$14,005	$14,461	$14,755	*$16,400*
Repairs	$10,268	$6,647	$6,774	$10,498	$12,552	*$18,080*
Misc. Service	$11,631	$11,775	$10,658	$11,024	$17,642	*$33,879*
Machinery & Equipment	$0	$0	$0	$0	$0	*$0*
SUBTOTAL Meter Reading	**$269,932**	**$226,923**	**$287,316**	**$284,904**	**$271,769**	***$297,155***
Utility Collections Subtotal	***$776,207***	***$701,992***	***$857,235***	***$870,719***	***$845,069***	***$876,054***
Wastewater Treatment						
Administration						
Salaries & Wages	$227,101	$244,913	$233,140	$230,269	$194,469	*$239,414*
Benefits	$92,261	$65,692	$106,139	$117,855	$81,424	*$97,344*
Supplies	$10,736	$8,534	$9,661	$10,840	$9,744	*$10,800*
Repairs	$159	$176	$938	$259	$965	*$1,000*
Misc. Service	$503,315	$436,272	$485,579	$463,109	$432,811	*$500,565*
SUBTOTAL Administration	**$833,572**	**$755,587**	**$835,457**	**$822,332**	**$719,413**	***$849,123***
Operations						
Salaries & Wages	$167,756	$164,971	$185,907	$162,005	$178,804	*$176,800*
Benefits	$76,648	$43,854	$91,666	$90,824	$82,000	*$78,081*
Supplies	$153,479	$178,520	$182,372	$172,080	$219,753	*$207,050*
Repairs	$0	$0	$0	$0	$0	*$0*
Misc. Service	$4,958	$1,504	$581	$5,490	$5,349	*$4,500*
SUBTOTAL Operations	**$402,841**	**$388,849**	**$460,526**	**$430,399**	**$485,906**	***$466,431***
Maintenance						
Salaries & Wages	$177,786	$199,391	$212,470	$276,289	$241,705	*$236,789*

Benefits	$100,930	$35,117	$111,570	$118,182	$96,614	*$107,851*
Supplies	$46,146	$47,396	$41,310	$44,327	$48,349	*$42,300*
Repairs	$181,359	$211,797	$252,356	$195,576	$184,695	*$180,500*
Misc. Service	$43,186	$43,367	$43,422	$43,560	$32,945	*$53,188*
SUBTOTAL Maintenance	**$549,407**	**$537,068**	**$661,128**	**$677,934**	**$604,308**	***$620,628***
Laboratory						
Salaries & Wages	$40,703	$54,727	$67,083	$72,268	$73,162	*$72,481*
Benefits	$24,741	$14,991	$50,731	$38,865	$33,890	*$33,347*
Supplies	$17,148	$27,954	$27,721	$20,981	$20,871	*$21,300*
Misc. Service	$117,742	$100,030	$122,482	$129,854	$91,776	*$113,600*
SUBTOTAL Laboratory	**$200,334**	**$197,702**	**$268,017**	**$261,968**	**$219,699**	***$240,728***
Wastewater Treatment Subtotal	***$1,986,154***	***$1,879,206***	***$2,225,128***	***$2,192,633***	***$2,029,326***	***$2,176,910***
Water Production						
Operations						
Salaries & Wages	$40,201	$42,737	$42,059	$46,211	$39,961	*$51,573*
Benefits	$16,928	$20,769	$31,587	$31,368	$28,062	*$26,522*
Supplies	$239,901	$228,120	$263,460	$299,170	$290,452	*$307,400*
Misc. Service						
Communications	$8,891	$8,761	$9,382	$10,639	$9,714	*$8,970*
Office/Equipment Rental	$1,378	$1,634	$1,918	$2,558	$604	*$905*
Special Services	$180,450	$184,556	$140,853	$193,163	$214,868	*$249,210*
Electric Service	$1,208,263	$1,242,564	$1,215,387	$1,196,382	$1,178,189	*$1,300,300*
Water Service	$0	$1,254	$1,183	$1,109	$1,650	*$1,500*
Subtotal Misc. Service	**$1,398,981**	**$1,438,769**	**$1,368,723**	**$1,403,851**	**$1,405,024**	***$1,560,885***
SUBTOTAL Operations	**$1,696,011**	**$1,730,395**	**$1,705,829**	**$1,780,600**	**$1,763,499**	***$1,946,380***
Maintenance						
Supplies	$9,277	$8,029	$12,454	$14,504	$11,893	*$17,500*
Repairs	$129,889	$138,929	$120,225	$140,349	$135,657	*$139,500*
Misc. Service	$2,930	$3,441	$3,441	$3,441	$3,441	*$3,441*
SUBTOTAL Maintenance	**$142,096**	**$150,399**	**$136,120**	**$158,294**	**$150,991**	***$160,441***
Water Production Subtotal	***$1,838,107***	***$1,880,794***	***$1,841,949***	***$1,938,894***	***$1,914,490***	***$2,106,821***
Water & Sewer Utilities						
Administration						
Salaries & Wages	$238,959	$210,552	$224,734	$249,718	$277,385	*$295,848*
Benefits	$106,527	$56,266	$114,764	$122,519	$118,470	*$126,841*
Supplies	$25,444	$25,078	$22,848	$21,958	$21,903	*$24,470*
Repairs	$13,340	$16,836	$12,952	$13,786	$25,888	*$21,180*

Misc. Service	$83,270	$89,657	$73,318	$75,598	$73,730	*$86,370*
SUBTOTAL Administration	**$467,540**	**$398,389**	**$448,616**	**$483,579**	**$517,376**	***$554,709***
Water Utilities						
Salaries & Wages	$286,796	$421,698	$404,337	$380,271	$443,119	*$510,409*
Benefits	$195,352	$88,257	$215,932	$222,920	$184,552	*$237,338*
Supplies	$92,377	$69,303	$63,923	$58,972	$73,251	*$67,510*
Repairs	$446,232	$447,649	$531,926	$524,161	$529,994	*$533,130*
Misc. Service	$125,474	$66,470	$53,889	$53,444	$35,109	*$57,998*
SUBTOTAL Water Utilities	**$1,146,231**	**$1,093,377**	**$1,270,007**	**$1,239,768**	**$1,266,025**	***$1,406,385***
Sewer Utilities						
Salaries & Wages	$469,586	$483,880	$562,211	$607,912	$607,210	*$575,963*
Benefits	$268,395	$106,766	$252,055	$290,893	$257,574	*$257,472*
Supplies	$213,928	$175,791	$202,794	$173,933	$164,570	*$180,980*
Repairs	$381,946	$399,199	$368,937	$360,345	$369,171	*$375,205*
Misc. Service	$277,551	$216,376	$209,122	$219,784	$158,357	*$263,433*
Machinery and Equipment	$6,922	$0	$0	$0	$0	*$0*
SUBTOTAL Sewer Utilities	**$1,618,328**	**$1,382,012**	**$1,595,119**	**$1,652,867**	**$1,556,882**	***$1,653,053***
Water & Sewer Utilities Subtotal	***$3,232,099***	***$2,873,778***	***$3,313,742***	***$3,376,214***	***$3,340,283***	***$3,614,147***
Non-Departmental						
Administration						
Salaries & Wages	$0	$0	$0	$0	$0	*$87,570*
Benefits	$299	$2,256	-$1,420	$2,916	$1,461	*$0*
Supplies	$0	$0	$0	$0	$866	*$996*
Repairs	$0	$0	$0	$0	$0	*$28,462*
Misc. Service	$129,633	$132,244	$146,311	$299,273	$271,288	*$274,477*
Sundry Charges						
Judgments and Damages	$0	$0	$0	$19,833	$0	*$0*
Bad Debts	$238,084	$159,124	$408,523	$82,849	$0	*$150,000*
G&A Charges	$2,224,072	$2,310,725	$2,385,669	$2,440,991	$2,641,151	*$2,788,422*
Funded Depreciation	$2,007,130	$2,056,085	$2,110,278	$2,110,278	$2,107,905	*$2,079,977*
Depreciation Expense	$2,359,022	$2,356,001	$2,355,686	$2,387,519	$0	*$0*
Depr Expense-Prior YR Adj	$0	$0	$18,905	$0	$0	*$0*
Subtotal Sundry Charges	**$6,828,308**	**$6,881,935**	**$7,279,061**	**$7,041,470**	**$4,749,056**	***$5,018,399***
Debt Service	$0	$0	$0	$0	$7,699	*$0*
Transfers						
General Fund	$409,000	$100,000	$368,928	$565,518	$100,000	*$100,000*
Group Health Insurance	$500,000	$0	$0	$0	$0	*$0*
GF Construction Fund	$0	$0	$84,917	$0	$62,500	*$0*

GO Debt Service Fund	$3,431,981	$3,771,795	$3,129,448	$3,778,005	$3,546,645	*$3,759,552*
Rev Bond Debt Svc Fund	$19,748	$0	$0	$0	$0	*$0*
Subtotal Transfers	**$4,360,729**	**$3,871,795**	**$3,583,293**	**$4,343,523**	**$3,709,145**	***$3,859,552***
Machinery and Equipment	$0	$0	$0	$0	$30,993	*$0*
Fixed Asset	-$1,898,258	-$841,338	-$2,430,643	-$4,860,833	$0	*$0*
Non-Departmental Subtotal	***$9,420,711***	***$10,046,892***	***$8,576,602***	***$6,826,349***	***$8,770,508***	***$9,269,456***
Total Operating Expenses	**$17,253,278**	**$17,382,662**	**$16,814,656**	**$15,204,809**	**$16,899,676**	**$18,043,388**
FUTURE DEBT SERVICE						
0	$0	$0	$0	$0	$0	$0
0	$0	$0	$0	$0	$0	$0
0	$0	$0	$0	$0	$0	$0
0	$0	$0	$0	$0	$0	$0
FUTURE DEBT SERVICE Subtotal	***$0***	***$0***	***$0***	***$0***	***$0***	***$0***
Total Expenses (with Depreciation)	**$17,253,278**	**$17,382,662**	**$16,814,656**	**$15,204,809**	**$16,899,676**	**$18,043,388**

Table A-2: Projected Expenses for the Water and Wastewater Fund, by Detailed Category

	Budget	Projected Expenses by Fiscal Year									
	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29
Projected Water Connections	*15,780*	15,938	16,097	16,258	16,421	16,585	16,750	16,918	17,087	17,258	17,431
Projected Water Production (MG)	*2,702*	2,729	2,756	2,784	2,812	2,840	2,868	2,897	2,926	2,955	2,985
Projected Metered Water Usage (MG)	*2,189*	2,210	2,232	2,255	2,278	2,300	2,323	2,347	2,370	2,394	2,418
Projected Metered Water Sales (MG)	*2,189*	2,210	2,232	2,255	2,278	2,300	2,323	2,347	2,370	2,394	2,418
Projected Wastewater Flows (MG)	*2,023*	2,043	2,063	2,084	2,105	2,126	2,147	2,169	2,191	2,212	2,235
Utility Collections											
Billing & Collections											
Salaries & Wages	*$261,419*	$270,569	$280,039	$289,840	$299,984	$310,483	$321,350	$332,597	$344,238	$356,286	$368,756
Benefits	*$117,765*	$121,887	$126,153	$130,568	$135,138	$139,868	$144,763	$149,830	$155,074	$160,502	$166,120
Supplies	*$79,760*	$81,355	$82,982	$84,642	$86,335	$88,062	$89,823	$91,619	$93,451	$95,320	$97,226
Repairs	*$9,330*	$10,263	$11,289	$12,418	$13,660	$15,026	$16,529	$18,182	$20,000	$22,000	$24,200
Misc. Service	*$110,625*	$120,581	$131,433	$143,262	$156,156	$170,210	$185,529	$202,227	$220,427	$240,265	$261,889
Machinery & Equipment	*$0*	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
SUBTOTAL Billing & Collections	***$578,899***	**$604,655**	**$631,896**	**$660,730**	**$691,273**	**$723,649**	**$757,994**	**$794,455**	**$833,190**	**$874,373**	**$918,191**
Meter Reading											
Salaries & Wages	*$155,384*	$160,822	$166,451	$172,277	$178,307	$184,548	$191,007	$197,692	$204,611	$211,772	$219,184
Benefits	*$73,412*	$75,981	$78,640	$81,392	$84,241	$87,189	$90,241	$93,399	$96,668	$100,051	$103,553
Supplies	*$16,400*	$16,728	$17,063	$17,404	$17,752	$18,107	$18,469	$18,838	$19,215	$19,599	$19,991
Repairs	*$18,080*	$19,888	$21,877	$24,065	$26,472	$29,119	$32,031	$35,234	$38,757	$42,633	$46,896
Misc. Service	*$33,879*	$38,961	$44,805	$51,526	$59,255	$68,143	$78,364	$90,119	$103,637	$119,183	$137,060

	Budget	Projected Expenses by Fiscal Year									
	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29
Machinery & Equipment	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
SUBTOTAL Meter Reading	**$297,155**	**$312,380**	**$328,836**	**$346,664**	**$366,027**	**$387,106**	**$410,112**	**$435,282**	**$462,888**	**$493,238**	**$526,684**
Utility Collections Subtotal	***$876,054***	***$917,035***	***$960,732***	***$1,007,394***	***$1,057,300***	***$1,110,755***	***$1,168,106***	***$1,229,737***	***$1,296,078***	***$1,367,611***	***$1,444,875***
Wastewater Treatment											
Administration											
Salaries & Wages	$239,414	$247,793	$256,466	$265,442	$274,732	$284,348	$294,300	$304,601	$315,262	$326,296	$337,716
Benefits	$97,344	$100,751	$104,277	$107,927	$111,704	$115,614	$119,660	$123,848	$128,183	$132,669	$137,312
Supplies	$10,800	$11,016	$11,236	$11,461	$11,690	$11,924	$12,162	$12,405	$12,653	$12,906	$13,164
Repairs	$1,000	$1,100	$1,210	$1,331	$1,464	$1,610	$1,771	$1,948	$2,143	$2,357	$2,593
Misc. Service	$500,565	$510,576	$520,788	$531,204	$541,828	$552,665	$563,718	$574,992	$586,492	$598,222	$610,186
SUBTOTAL Administration	**$849,123**	**$871,236**	**$893,977**	**$917,365**	**$941,418**	**$966,161**	**$991,611**	**$1,017,794**	**$1,044,733**	**$1,072,450**	**$1,100,971**
Operations											
Salaries & Wages	$176,800	$182,988	$189,393	$196,022	$202,883	$209,984	$217,333	$224,940	$232,813	$240,961	$249,395
Benefits	$78,081	$80,814	$83,642	$86,569	$89,599	$92,735	$95,981	$99,340	$102,817	$106,416	$110,141
Supplies	$207,050	$221,544	$237,052	$253,646	$271,401	$290,399	$310,727	$332,478	$355,751	$380,654	$407,300
Repairs	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Misc. Service	$4,500	$4,590	$4,682	$4,776	$4,872	$4,969	$5,068	$5,169	$5,272	$5,377	$5,485
SUBTOTAL Operations	**$466,431**	**$489,936**	**$514,769**	**$541,013**	**$568,755**	**$598,087**	**$629,109**	**$661,927**	**$696,653**	**$733,408**	**$772,321**
Maintenance											
Salaries & Wages	$236,789	$245,077	$253,655	$262,533	$271,722	$281,232	$291,075	$301,263	$311,807	$322,720	$334,015
Benefits	$107,851	$111,626	$115,533	$119,577	$123,762	$128,094	$132,577	$137,217	$142,020	$146,991	$152,136
Supplies	$42,300	$43,146	$44,009	$44,889	$45,787	$46,703	$47,637	$48,590	$49,562	$50,553	$51,564
Repairs	$180,500	$204,000	$208,080	$212,242	$216,487	$220,817	$225,233	$229,738	$234,333	$239,020	$243,800
Misc. Service	$53,188	$54,252	$55,337	$56,444	$57,573	$58,724	$59,898	$61,096	$62,318	$63,564	$64,835
SUBTOTAL Maintenance	$620,628	$658,101	$681,614	$700,785	$720,533	$740,876	$761,832	$783,424	$805,670	$828,591	$852,208

	Budget	Projected Expenses by Fiscal Year									
	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29
Laboratory											
Salaries & Wages	*$72,481*	$75,018	$77,644	$80,362	$83,175	$86,086	$89,099	$92,217	$95,445	$98,786	$102,244
Benefits	*$33,347*	$34,514	$35,722	$36,972	$38,266	$39,605	$40,991	$42,426	$43,911	$45,448	$47,039
Supplies	*$21,300*	$21,726	$22,161	$22,604	$23,056	$23,517	$23,987	$24,467	$24,956	$25,455	$25,964
Misc. Service	***$113,600***	**$115,872**	**$118,189**	**$120,553**	**$122,964**	**$125,423**	**$127,931**	**$130,490**	**$133,100**	**$135,762**	**$138,477**
SUBTOTAL Laboratory	***$240,728***	***$247,130***	***$253,716***	***$260,491***	***$267,461***	***$274,631***	***$282,008***	***$289,600***	***$297,412***	***$305,451***	***$313,724***
Wastewater Treatment Subtotal	**$2,176,910**	**$2,266,403**	**$2,344,076**	**$2,419,654**	**$2,498,167**	**$2,579,755**	**$2,664,560**	**$2,752,745**	**$2,844,468**	**$2,939,900**	**$3,039,224**
Water Production											
Operations											
Salaries & Wages	*$51,573*	$53,378	$55,246	$57,180	$59,181	$61,252	$63,396	$65,615	$67,912	$70,289	$72,749
Benefits	*$26,522*	$27,450	$28,411	$29,405	$30,434	$31,499	$32,601	$33,742	$34,923	$36,145	$37,410
Supplies	*$307,400*	$322,770	$338,909	$355,854	$373,647	$392,329	$411,945	$432,542	$454,169	$476,877	$500,721
Misc. Service											
Communications	*$8,970*	$9,149	$9,332	$9,519	$9,709	$9,903	$10,101	$10,303	$10,509	$10,719	$10,933
Office/Equipment Rental	*$905*	$968	$1,036	$1,109	$1,187	$1,270	$1,359	$1,454	$1,556	$1,665	$1,782
Special Services	*$249,210*	$209,000	$229,900	$252,890	$278,179	$305,997	$336,597	$370,257	$407,283	$448,011	$492,812
Electric Service	*$1,300,300*	$1,233,508	$1,266,813	$1,301,361	$1,336,715	$1,372,893	$1,409,914	$1,448,295	$1,487,571	$1,527,763	$1,569,415
Water Service	*$1,500*	$1,605	$1,717	$1,837	$1,966	$2,104	$2,251	$2,409	$2,578	$2,758	$2,951
Subtotal Misc. Service	***$1,560,885***	**$1,454,230**	**$1,508,798**	**$1,566,716**	**$1,627,756**	**$1,692,167**	**$1,760,222**	**$1,832,718**	**$1,909,497**	**$1,990,916**	**$2,077,893**
SUBTOTAL Operations	***$1,946,380***	**$1,857,828**	**$1,931,364**	**$2,009,155**	**$2,091,018**	**$2,177,247**	**$2,268,164**	**$2,364,617**	**$2,466,501**	**$2,574,227**	**$2,688,773**
Maintenance											
Supplies	*$17,500*	$13,645	$13,780	$13,920	$14,060	$14,200	$14,340	$14,485	$14,630	$14,775	$14,925
Repairs	*$139,500*	$136,450	$137,800	$139,200	$140,600	$142,000	$143,400	$144,850	$146,300	$147,750	$149,250
Misc. Service	*$3,441*	$3,510	$3,580	$3,652	$3,725	$3,800	$3,876	$3,954	$4,033	$4,114	$4,196

	Budget	Projected Expenses by Fiscal Year									
	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29
SUBTOTAL Maintenance	***$160,441***	**$153,605**	**$155,160**	**$156,772**	**$158,385**	**$160,000**	**$161,616**	**$163,289**	**$164,963**	**$166,639**	**$168,371**
Water Production Subtotal	***$2,106,821***	***$2,011,433***	***$2,086,524***	***$2,165,927***	***$2,249,403***	***$2,337,247***	***$2,429,780***	***$2,527,906***	***$2,631,464***	***$2,740,866***	***$2,857,144***
Water & Sewer Utilities											
Administration											
Salaries & Wages	*$295,848*	$306,203	$316,920	$328,012	$339,492	$351,374	$363,672	$376,401	$389,575	$403,210	$417,322
Benefits	*$126,841*	$131,280	$135,875	$140,631	$145,553	$150,647	$155,920	$161,377	$167,025	$172,871	$178,921
Supplies	*$24,470*	$24,959	$25,458	$25,967	$26,486	$27,016	$27,556	$28,107	$28,669	$29,242	$29,827
Repairs	*$21,180*	$22,874	$24,704	$26,680	$28,814	$31,119	$33,609	$36,298	$39,202	$42,338	$45,725
Misc. Service	*$86,370*	$88,097	$89,859	$91,656	$93,489	$95,359	$97,266	$99,211	$101,195	$103,219	$105,283
SUBTOTAL Administration	*$554,709*	$573,413	$592,816	$612,946	$633,834	$655,515	$678,023	$701,394	$725,666	$750,880	$777,078
Water Utilities											
Salaries & Wages	*$510,409*	$528,273	$546,763	$565,900	$585,707	$606,207	$627,424	$649,384	$672,112	$695,636	$719,983
Benefits	*$237,338*	$245,645	$254,243	$263,142	$272,352	$281,884	$291,750	$301,961	$312,530	$323,469	$334,790
Supplies	*$67,510*	$68,860	$70,237	$71,642	$73,075	$74,537	$76,028	$77,549	$79,100	$80,682	$82,296
Repairs	*$533,130*	$507,594	$512,616	$517,824	$523,032	$528,240	$533,448	$538,842	$544,236	$549,630	$570,523
Misc. Service	*$57,998*	$59,158	$60,341	$61,548	$62,779	$64,035	$65,316	$66,622	$67,954	$69,313	$70,699
SUBTOTAL Water Utilities	*$1,406,385*	$1,409,530	$1,444,200	$1,480,056	$1,516,945	$1,554,903	$1,593,966	$1,634,358	$1,675,932	$1,718,730	$1,778,291
Sewer Utilities											
Salaries & Wages	*$575,963*	$596,122	$616,986	$638,581	$660,931	$684,064	$708,006	$732,786	$758,434	$784,979	$812,453
Benefits	*$257,472*	$266,484	$275,811	$285,464	$295,455	$305,796	$316,499	$327,576	$339,041	$350,907	$363,189
Supplies	*$180,980*	$193,800	$197,676	$201,630	$205,663	$209,776	$213,972	$218,251	$222,616	$227,068	$231,609
Repairs	*$375,205*	$379,331	$383,084	$386,976	$390,868	$394,760	$398,652	$402,683	$406,714	$410,745	$418,960
Misc. Service	*$263,433*	$268,702	$274,076	$279,558	$285,149	$290,852	$296,669	$302,602	$308,654	$314,827	$321,124
Machinery and Equipment	***$0***	**$0**	**$0**	**$0**	**$0**	**$0**	**$0**	**$0**	**$0**	**$0**	**$0**
SUBTOTAL Sewer Utilities	***$1,653,053***	***$1,704,439***	***$1,747,633***	***$1,792,209***	***$1,838,066***	***$1,885,248***	***$1,933,798***	***$1,983,898***	***$2,035,459***	***$2,088,526***	***$2,147,335***

	Budget	Projected Expenses by Fiscal Year									
	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29
Water & Sewer Utilities Subtotal	***$3,614,147***	***$3,687,382***	***$3,784,649***	***$3,885,211***	***$3,988,845***	***$4,095,666***	***$4,205,787***	***$4,319,650***	***$4,437,057***	***$4,558,136***	***$4,702,704***
Non-Departmental											
Administration											
Salaries & Wages	*$87,570*	$90,635	$93,807	$97,090	$100,488	$104,005	$107,645	$111,413	$115,312	$119,348	$123,525
Benefits	*$0*	$1,035	$1,071	$1,108	$1,147	$1,187	$1,229	$1,272	$1,317	$1,363	$1,411
Supplies	*$996*	$1,056	$1,119	$1,186	$1,257	$1,332	$1,412	$1,497	$1,587	$1,682	$1,783
Repairs	*$28,462*	$29,031	$29,612	$30,204	$30,808	$31,424	$32,052	$32,693	$33,347	$34,014	$34,694
Misc. Service	*$274,477*	$296,435	$320,150	$345,762	$373,423	$403,297	$435,561	$470,406	$508,038	$548,681	$592,575
Sundry Charges											
Judgments and Damages	*$0*	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Bad Debts	*$150,000*	$153,000	$156,060	$159,181	$162,365	$165,612	$168,924	$172,302	$175,748	$179,263	$182,848
G&A Charges	*$2,788,422*	$2,913,901	$3,045,027	$3,182,053	$3,325,245	$3,474,881	$3,631,251	$3,794,657	$3,965,417	$4,143,861	$4,330,335
Funded Depreciation	*$2,079,977*	$2,100,777	$2,121,785	$2,143,003	$2,164,433	$2,186,077	$2,207,938	$2,230,017	$2,252,317	$2,274,840	$2,297,588
Depreciation Expense	*$0*	$2,407,200	$2,455,344	$2,504,451	$2,554,540	$2,605,631	$2,657,744	$2,710,899	$2,765,117	$2,820,419	$2,876,827
Depr Expense-Prior YR Adj	***$0***	**$0**	**$0**	**$0**	**$0**	**$0**	**$0**	**$0**	**$0**	**$0**	**$0**
Subtotal Sundry Charges	*$5,018,399*	$7,574,878	$7,778,216	$7,988,688	$8,206,583	$8,432,201	$8,665,857	$8,907,875	$9,158,599	$9,418,383	$9,687,598
Debt Service	*$0*	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Transfers											
General Fund	*$100,000*	$101,000	$102,010	$103,030	$104,060	$105,101	$106,152	$107,214	$108,286	$109,369	$110,463
Group Health Insurance	*$0*	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
GF Construction Fund	*$0*	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
GO Debt Service Fund	*$3,759,552*	$3,578,752	$3,647,327	$2,567,677	$2,351,402	$2,341,515	$2,294,558	$2,125,735	$3,142,793	$2,993,555	$3,299,360

	Budget	Projected Expenses by Fiscal Year									
	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29
Rev Bond Debt Svc Fund	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Subtotal Transfers	$3,859,552	$3,679,752	$3,749,337	$2,670,707	$2,455,462	$2,446,616	$2,400,710	$2,232,949	$3,251,079	$3,102,924	$3,409,823
Machinery and Equipment	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Fixed Asset	$0	-$2,560,200	-$2,611,404	-$2,663,632	-$2,716,905	-$2,771,243	-$2,826,668	-$2,883,201	-$2,940,865	-$2,999,682	-$3,059,676
Non-Departmental Subtotal	**$9,269,456**	**$9,112,622**	**$9,361,908**	**$8,471,113**	**$8,452,263**	**$8,648,819**	**$8,817,798**	**$8,874,904**	**$10,128,414**	**$10,226,713**	**$10,791,733**
Total Operating Expenses	**$18,043,388**	**$17,996,684**	**$18,540,950**	**$17,952,949**	**$18,249,433**	**$18,774,580**	**$19,286,178**	**$19,701,651**	**$21,329,311**	**$21,818,520**	**$22,812,539**
FUTURE DEBT SERVICE	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29
FY2020 Series $7.5M			$514,200	$514,200	$514,200	$514,200	$514,200	$514,200	$514,200	$514,200	$514,200
FY2022 Series $7.5M					*$534,974*	*$534,974*	*$534,974*	*$534,974*	*$534,974*	*$534,974*	*$534,974*
0% Coverage Required on NEW Debt		$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
FUTURE *DEBT SERVICE* Subtotal	**$0**	**$0**	**$514,200**	**$514,200**	**$1,049,174**	**$1,049,174**	**$1,049,174**	**$1,049,174**	**$1,049,174**	**$1,049,174**	**$1,049,174**
TOTAL EXPENSES (with depreciation)	***$18,043,388***	***$17,994,875***	***$19,052,089***	***$18,463,499***	***$19,295,152***	***$19,821,416***	***$20,335,205***	***$20,754,116***	***$22,386,655***	***$22,882,400***	***$23,884,854***
Projected REVENUE at current rates (From Table 4-1)	***$18,061,808***	**$18,106,074**	**$18,277,058**	**$18,455,269**	**$18,637,494**	**$18,809,954**	**$18,989,644**	**$19,176,419**	**$19,360,558**	**$19,548,347**	**$19,736,570**
Deficit (-) or Surplus (+)	*$18,420*	*$111,199*	*-$775,031*	*-$8,230*	*-$657,658*	*-$1,011,462*	*-$1,345,561*	*-$1,577,697*	*-$3,026,097*	*-$3,334,053*	*-$4,148,284*

Table A-3: Basis for Projections of Expenses for the Water and Wastewater Fund

Expense Category	Basis for projections
Utility Collections	
Billing & Collections	
Salaries & Wages	Begin w/ FY2019 budget, inflate at 3.5%/yr
Benefits	Begin w/ FY2019 budget, inflate at 3.5%/yr
Supplies	Begin w/ FY2019 budget, inflate at 2.0%/yr
Repairs	Begin w/ FY2019 budget, inflate at 10.0%/yr
Misc. Service	Begin w/ FY2019 budget, inflate at 9.0%/yr
Machinery & Equipment	Remain at $0
Meter Reading	
Salaries & Wages	Begin w/ FY2019 budget, inflate at 3.5%/yr
Benefits	Begin w/ FY2019 budget, inflate at 3.5%/yr
Supplies	Begin w/ FY2019 budget, inflate at 2.0%/yr
Repairs	Begin w/ FY2019 budget, inflate at 10.0%/yr
Misc. Service	Begin w/ FY2019 budget, inflate at 15.0%/yr
Machinery & Equipment	Remain at $0
Wastewater Treatment	
Administration	
Salaries & Wages	Begin w/ FY2019 budget, inflate at 3.5%/yr
Benefits	Begin w/ FY2019 budget, inflate at 3.5%/yr
Supplies	Begin w/ FY2019 budget, inflate at 2.0%/yr
Repairs	Begin w/ FY2019 budget, inflate at 10.0%/yr
Misc. Service	Begin w/ FY2019 budget, inflate at 2.0%/yr
Operations	
Salaries & Wages	Begin w/ FY2019 budget, inflate at 3.5%/yr
Benefits	Begin w/ FY2019 budget, inflate at 3.5%/yr
Supplies	Begin w/ FY2019 budget, inflate at 7.0%/yr
Repairs	Remain at $0
Misc. Service	Begin w/ FY2019 budget, inflate at 2.0%/yr
Maintenance	
Salaries & Wages	Begin w/ FY2019 budget, inflate at 3.5%/yr
Benefits	Begin w/ FY2019 budget, inflate at 3.5%/yr
Supplies	Begin w/ FY2019 budget, inflate at 2.0%/yr
Repairs	Begin at $200,000, inflate at 2.0%/yr
Misc. Service	Begin w/ FY2019 budget, inflate at 2.0%/yr

Laboratory	
Salaries & Wages	Begin w/ FY2019 budget, inflate at 3.5%/yr
Benefits	Begin w/ FY2019 budget, inflate at 3.5%/yr
Supplies	Begin w/ FY2019 budget, inflate at 2.0%/yr
Misc. Service	Begin w/ FY2019 budget, inflate at 2.0%/yr
Water Production	
Operations	
Salaries & Wages	Begin w/ FY2019 budget, inflate at 3.5%/yr
Benefits	Begin w/ FY2019 budget, inflate at 3.5%/yr
Supplies	Begin w/ FY2019 budget, inflate at 5.0%/yr
Misc. Service	
Communications	Begin w/ FY2019 budget, inflate at 2.0%/yr
Office/Equipment Rental	Begin w/ FY2019 budget, inflate at 7.0%/yr
Special Services	Begin at $190,000, inflate at 10.0%/yr
Electric Service	Based on $452 per MG of water produced, inflated at 1.7%/yr
Water Service	Begin w/ FY2019 budget, inflate at 7.0%/yr
Maintenance	
Supplies	Based on $5 per MG of water produced, inflated at 2.0%/yr
Repairs	Based on $50 per MG of water produced, inflated at 2.2%/yr
Misc. Service	Begin w/ FY2019 budget, inflate at 2.0%/yr
Water & Sewer Utilities	
Administration	
Salaries & Wages	Begin w/ FY2019 budget, inflate at 3.5%/yr
Benefits	Begin w/ FY2019 budget, inflate at 3.5%/yr
Supplies	Begin w/ FY2019 budget, inflate at 2.0%/yr
Repairs	Begin w/ FY2019 budget, inflate at 8.0%/yr
Misc. Service	Begin w/ FY2019 budget, inflate at 2.0%/yr
Water Utilities	
Salaries & Wages	Begin w/ FY2019 budget, inflate at 3.5%/yr
Benefits	Begin w/ FY2019 budget, inflate at 3.5%/yr
Supplies	Begin w/ FY2019 budget, inflate at 2.0%/yr
Repairs	Based on $186 per MG of water produced, inflated at 3.8%/yr
Misc. Service	Begin w/ FY2019 budget, inflate at 2.0%/yr
Sewer Utilities	

Salaries & Wages	Begin w/ FY2019 budget, inflate at 3.5%/yr
Benefits	Begin w/ FY2019 budget, inflate at 4%/yr
Supplies	Begin w/ 2015-2018 average, inflate at 2.0%/yr
Repairs	Based on $139 per MG of water produced, inflated at 2.0%/yr
Misc. Service	Begin w/ FY2019 budget, inflate at 2%/yr
Non-Departmental	
Administration	
Salaries & Wages	Begin w/ FY2019 budget, inflate at 3.5%/yr
Benefits	Begin with $1,000, inflate at 3.5%/yr
Supplies	Begin w/ FY2019 budget, inflate at 6.0%/yr
Repairs	Begin w/ FY2019 budget, inflate at 2.0%/yr
Misc. Service	Begin w/ FY2019 budget, inflate at 8.0%/yr
Sundry Charges	
Judgments and Damages	Remain at $0
Bad Debts	Begin w/ FY2019 budget, inflate at 2.0%/yr
G&A Charges	Begin w/ FY2019 budget, inflate at 4.5%/yr
Funded Depreciation	Begin w/ FY2019 budget, inflate at 1.0%/yr
Depreciation Expense	Begin w/ 2014-2017 average, inflate at 2.0%/yr. This represents the decrease in asset value because of aging.
Depr Expense-Prior YR Adj	Remain at $0
Subtotal Sundry Charges	
Debt Service	See below under Transfers to Debt Service Fund
Transfers	
General Fund	Begin w/ FY2019 budget, inflate at 1.0%/yr to match population growth
Group Health Insurance	Remain at $0
GF Construction Fund	Remain at $0
GO Debt Service Fund	Provided by city
Rev Bond Debt Svc Fund	Remain at $0
Machinery and Equipment	Remain at $0
Fixed Asset	Begin w/ 2014-2017 average, inflate at 2.0%/yr. This is an offset to depreciation, recognizing the increase in asset value because of improvements made.